



TADI campaign 2024

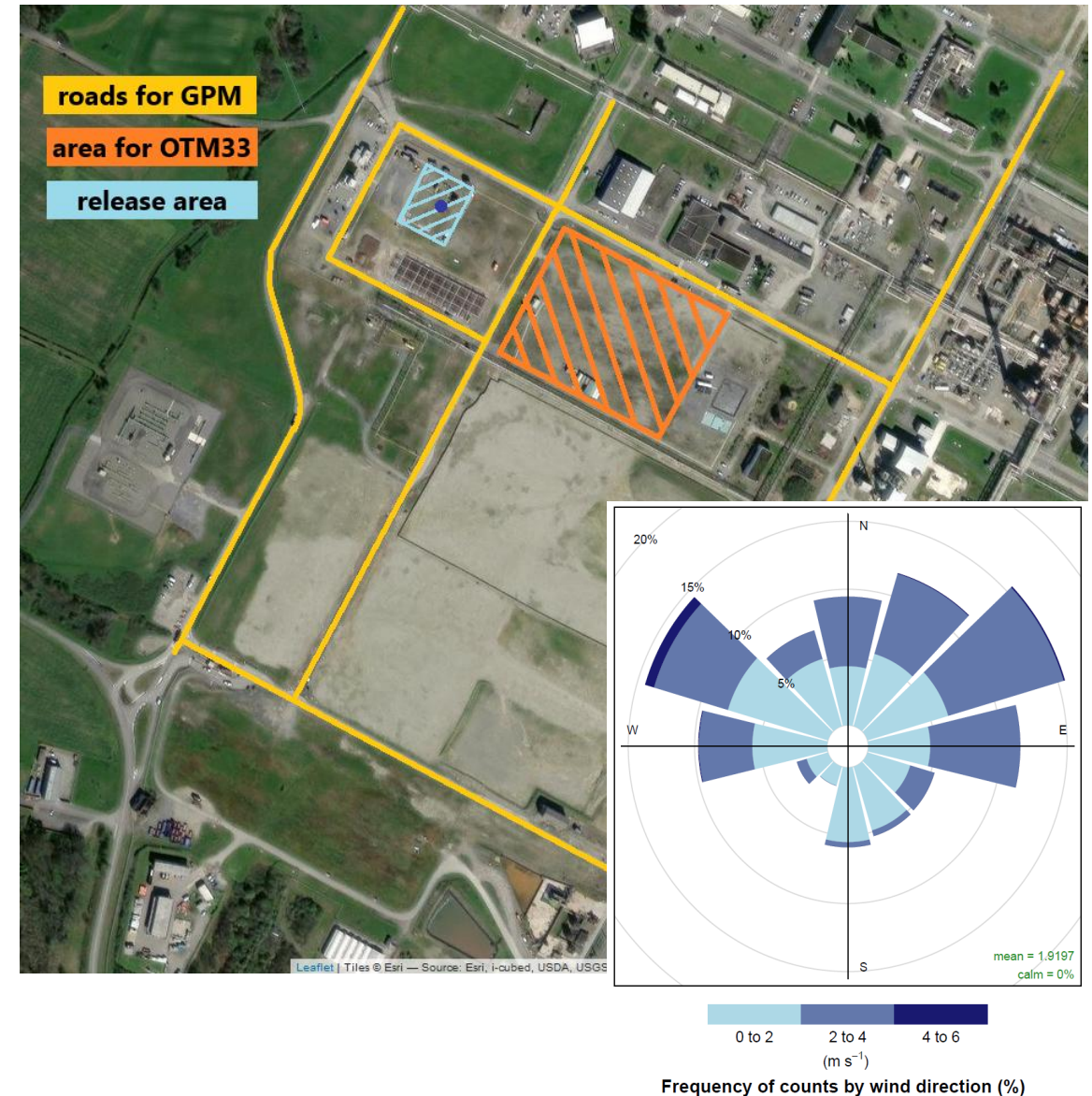
Controlled blind release experiment in Lacq, southern France

Heidelberg University

June 24 to 28

Set-Up

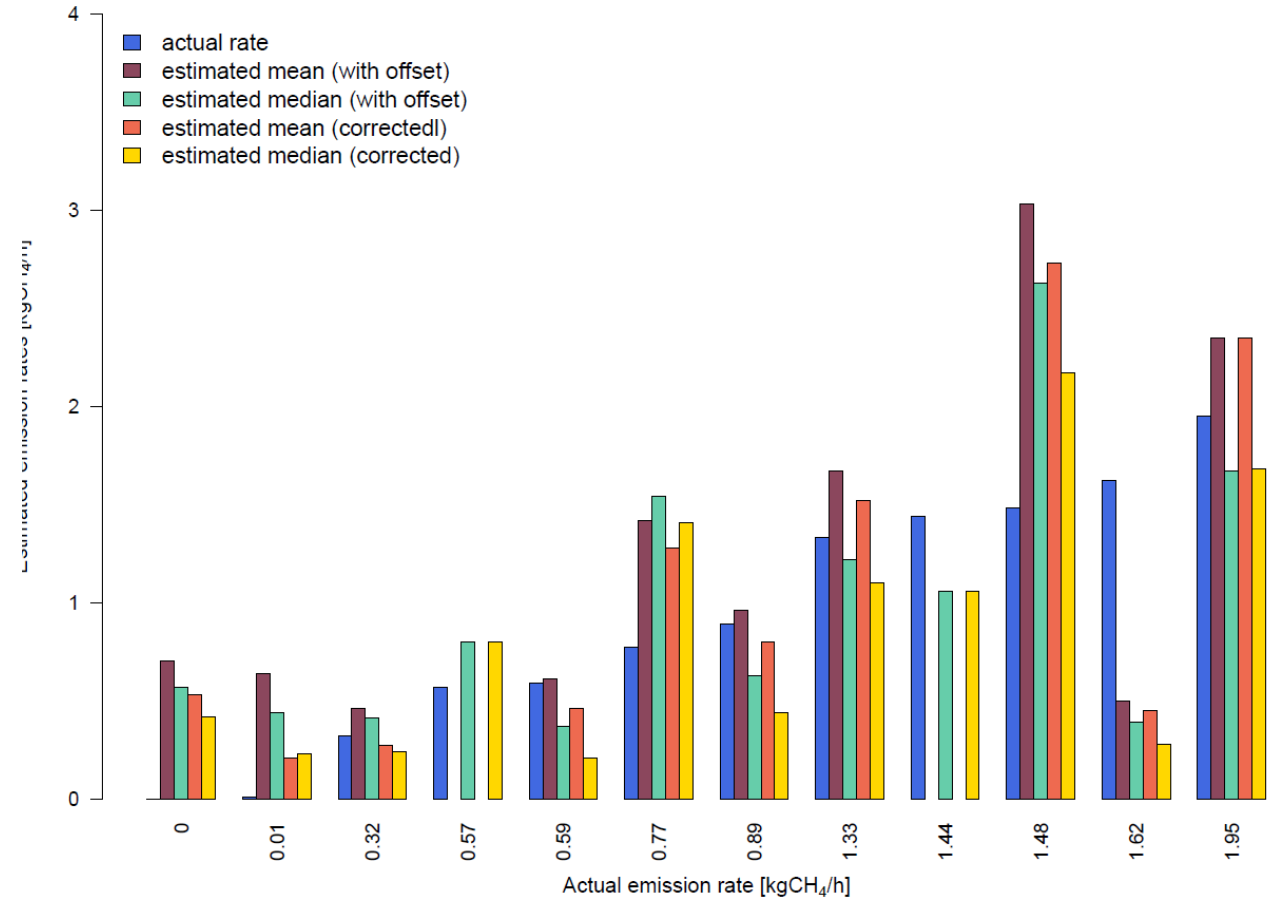
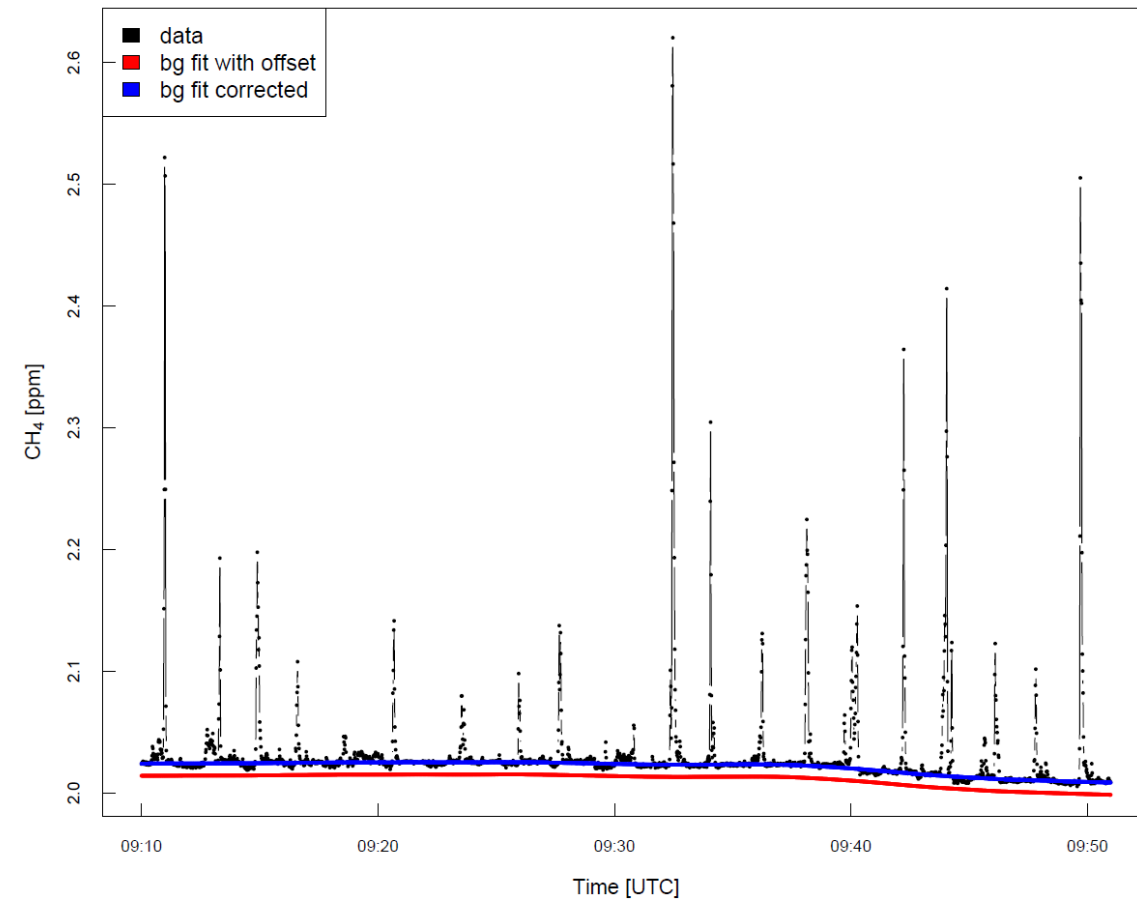
- June 24 to 28
- from 0 to 200 kgCH₄/h
- 45 minutes, with 5 to 10 minute breaks
- position of point source (height and location) varied over area of 40 m x 60 m and heights between 0 to 6.5 m a.g.
- measurements performed during 34 releases
- average wind speed of 1.9 m/s, mainly directing from the north
- temperature ranged from 18 to 33 °C
- measurements conducted with a driving speed of approximately 15 km/h
- number of plume crossing varied between 10 and 26 and sometimes performed at two different distances
- evaluation was done via a simple Gaussian plume model and in three cases with the OTM33-a method



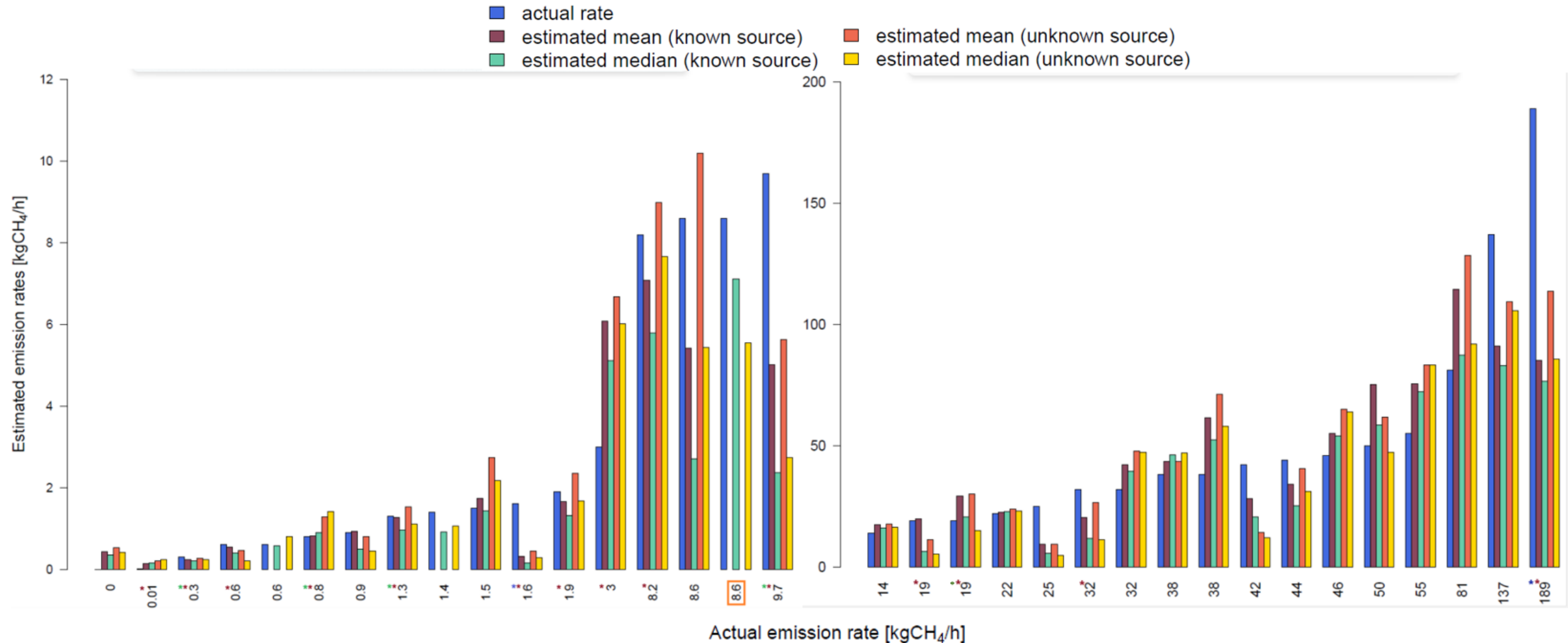
Correction of background calculation

- during the first analysis the background calculation was performed incorrectly (error in the programming code)
- constant CH₄ offset of 10 ppb was additionally subtracted from the calculated CH₄ background -> CH₄ enhancements (peaks) are 10 ppb too high
- background was corrected for all release experiments with release rates < 2 kgCH₄ h⁻¹ (35% of all releases) -> **see white paper**
- For the controlled CH₄ release experiments with release rates between 0.5 and 2.0 kgCH₄ h⁻¹ the release rates calculated and reported by Heidelberg University were estimated 10-30% too high due to the calculation error in the background of the CH₄ concentration

Correction of background calculation



Estimated emission rates for unknown and known source location



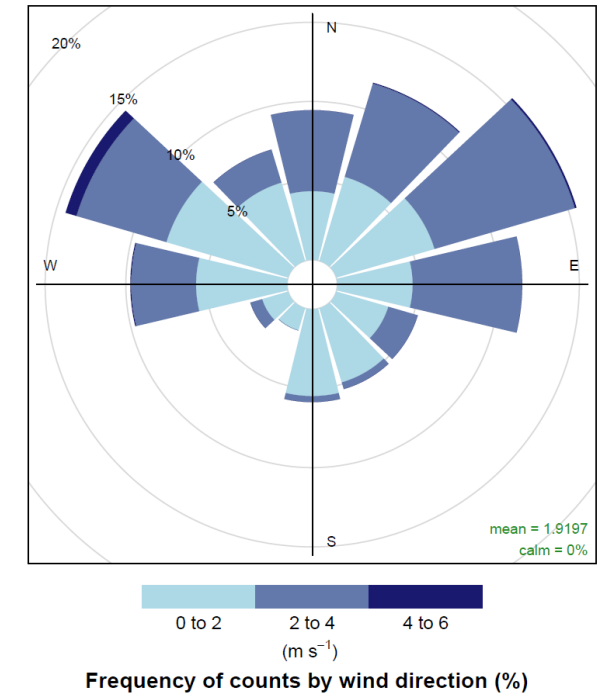
Comparison of the actual and estimated emissions rates of 34 CH₄ releases given in kgCH₄/h. Estimated rates are given as mean and median of the corresponding determined transect values. Red, green, blue stars near the actual emission rate indicate a wind speed below 2 m/s, 1.5 m/s, 1 m/s, respectively, and the orange square indicates a not accepted OTM.

Estimated emission rates for unknown and known source location

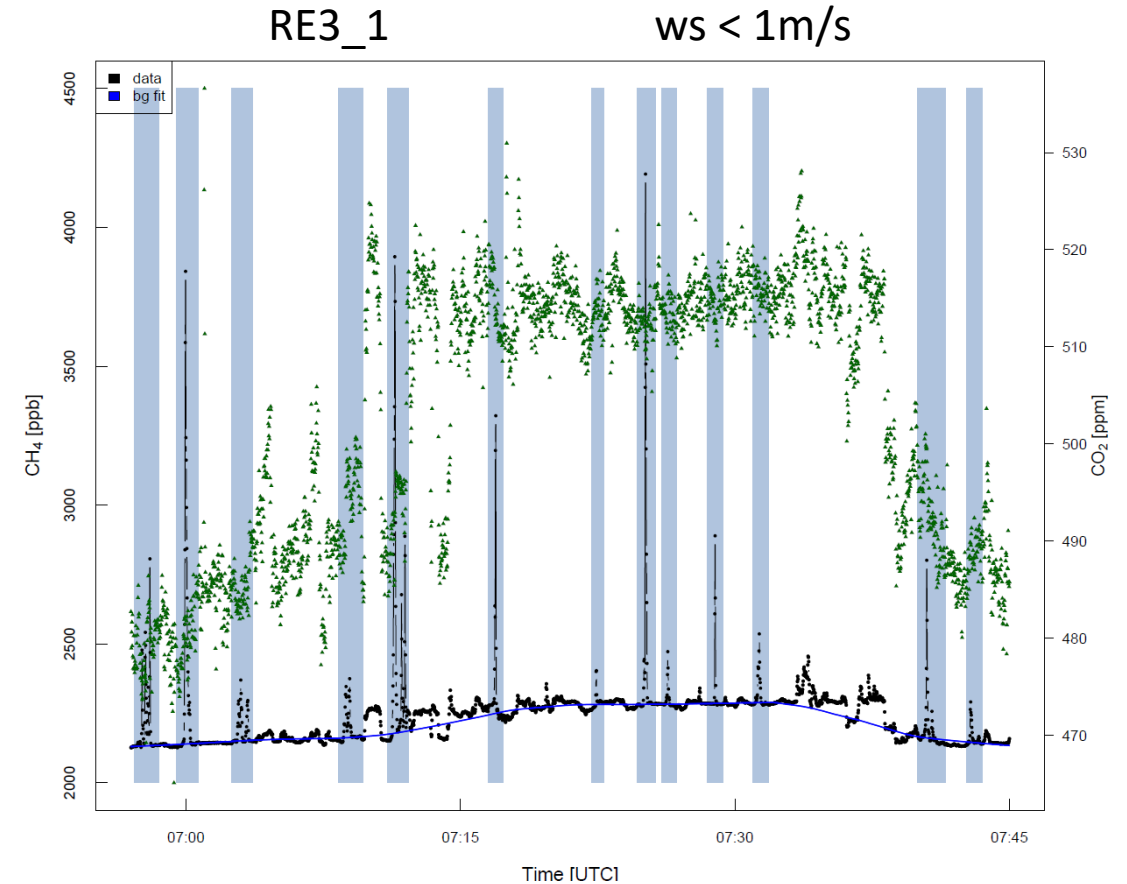
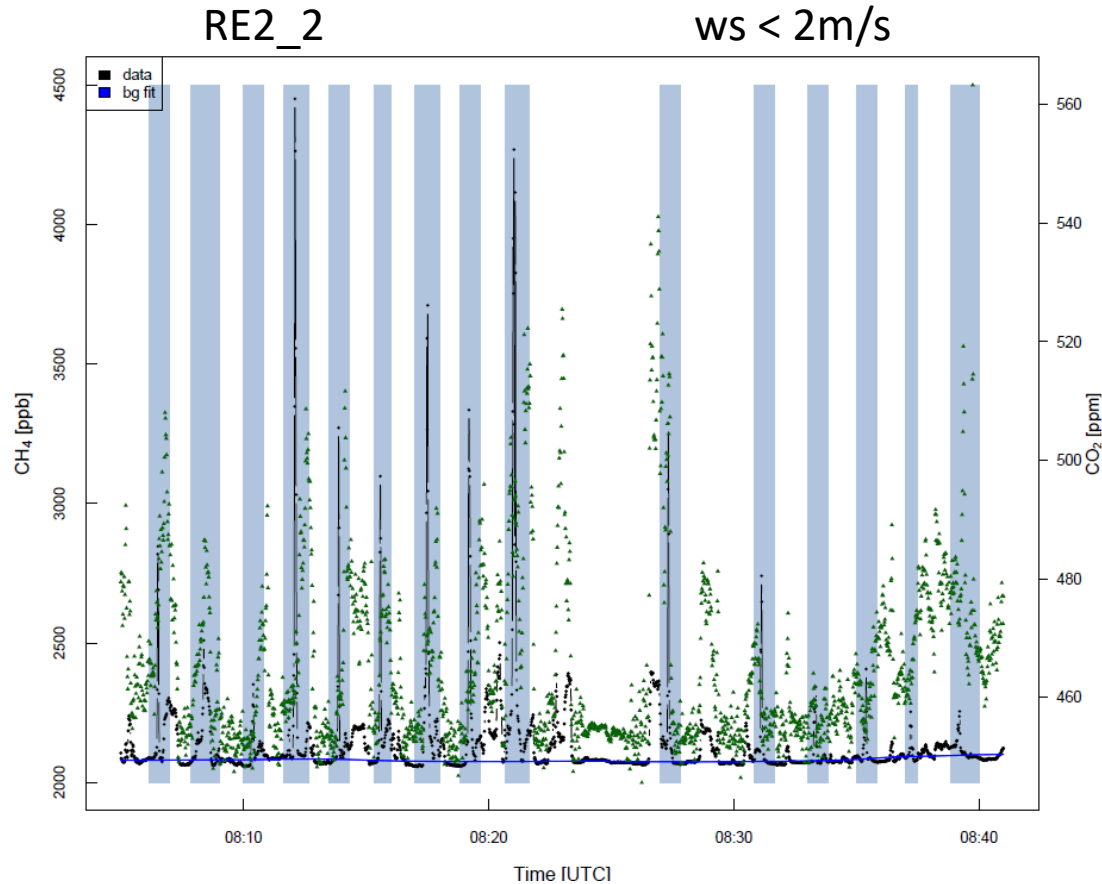
- Zero-release und 0.01-release are excluded
- with knowledge about true release position the distance changed only from 1 to maximum 29 meters (relative distance changed around 6% in average -> only small influence on determined emission rates)
- release height was assumed at 2 m a.g. while the actual height was between 0 and 6.5 m; the majority of releases were at heights between 1 and 3 m, which was close enough to our estimate not to detect any significant improvement after height adjustment
- cases of 6.5 m und 0 m height are improved around 20% in the mean estimated emission rate and 13% in the median after position adjustment
- nearly half of the estimated emission rates showed a slight improvement after position adjustment

Wind conditions

- 20 releases with wind speeds $>2\text{ m/s}$ (relative deviation: 32% (mean); 32% (median))
- 14 releases with wind speeds $< 2\text{ m/s}$ (35% (mean); 46% (median) (mainly underestimated))
- 7 releases with wind speeds $< 1.5\text{ m/s}$ (40% (mean); 45% (median) (mainly underestimated))
- 2 releases with wind speeds $< 1\text{ m/s}$ (68% (mean); 76% (median) (underestimated))



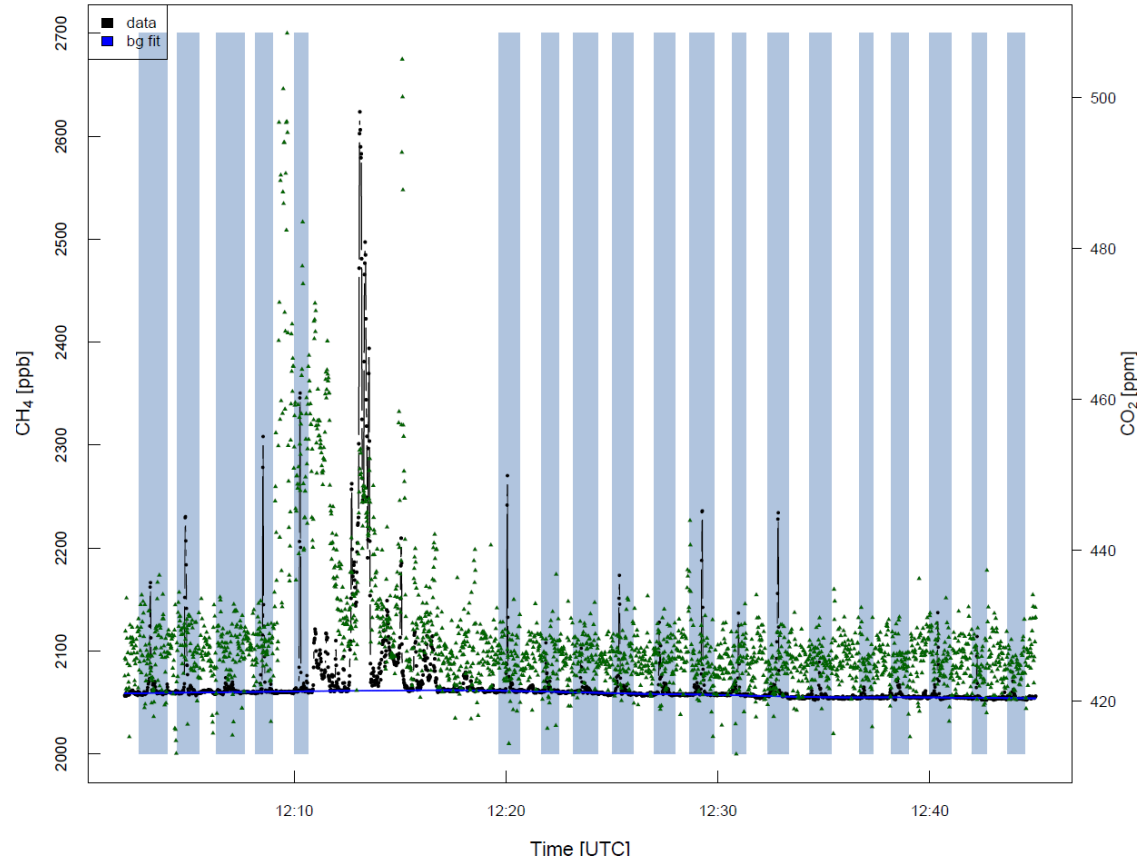
Influence of co-emitters



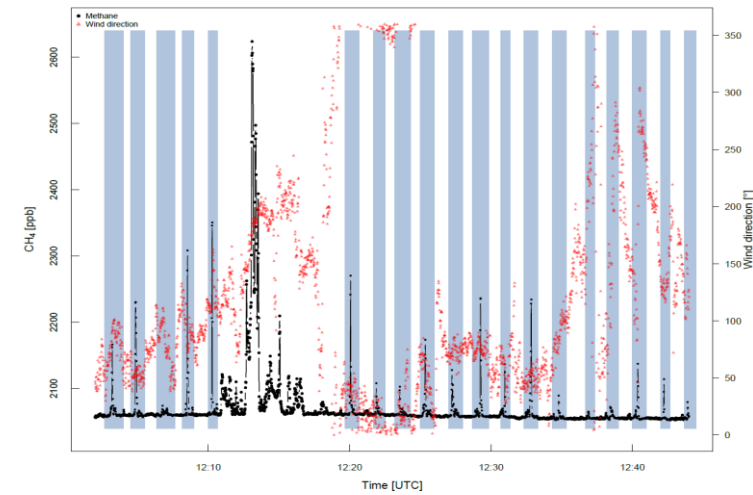
Variations in measured background can influence the emission rate estimation, especially at lower release rates ($< 2 \text{ kgCH}_4/\text{h}$); sometimes other emitters can influence the measurements as it can be seen in changing CO₂ values

0.01 kg/h-release (RE3_5)

Calculated emission rate: 0.21 ± 0.03 kgCH₄/h (mean)
 0.23 ± 0.06 kgCH₄/h (median)

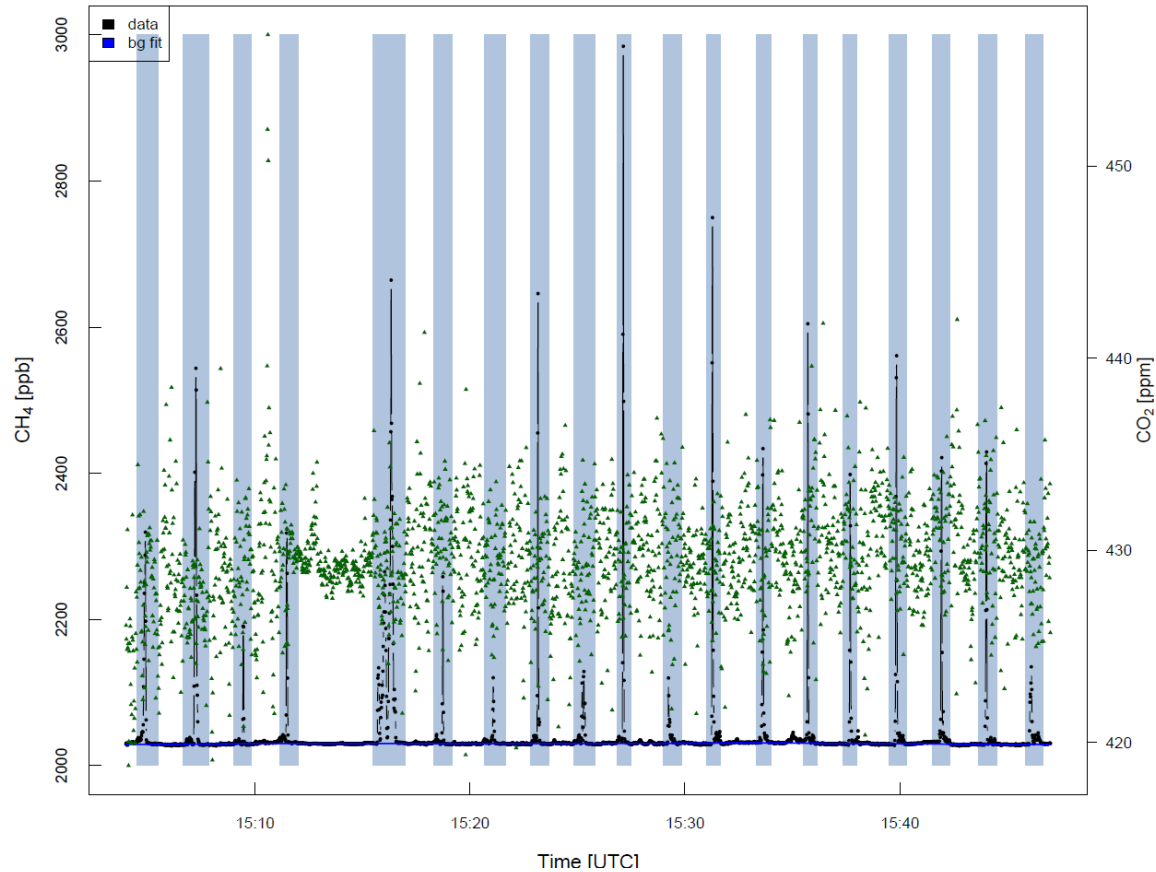


- Unstable wind conditions
- background (~2060 ppb)
- Peaks are consistent with wind direction
- Enhancement during rapid changing wd excluded
- Enhancements range from 10 to 290 ppb CH₄ above background

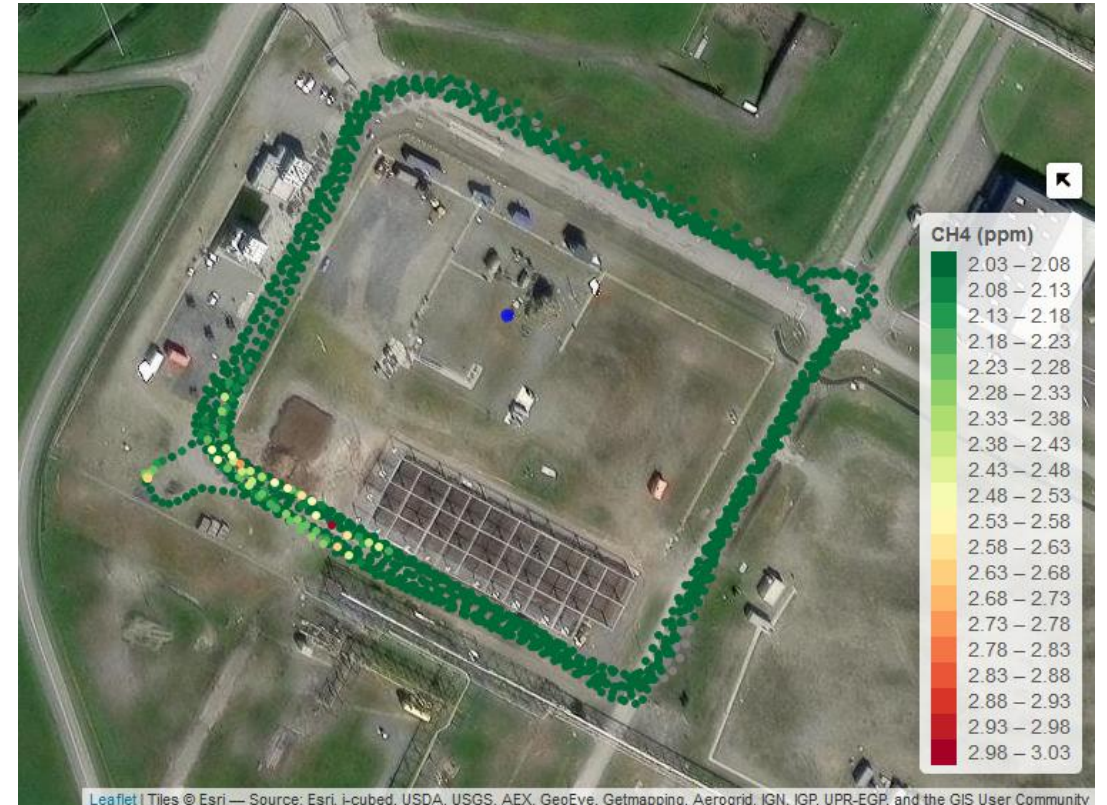


Zero-release (RE3_8)

Calculated emission rate: 0.53 ± 0.13 kgCH₄/h (mean)
 0.42 ± 0.15 kgCH₄/h (median)



- Stable wind conditions
- Clear background (~2030 ppb)
- Peaks are consistent with wind direction
- Enhancements range from 90 to 950 ppb CH₄ above background



Main Findings

- Position adjustment doesn't change the results significantly in most cases but the releases with a strong height adjustment are improved by 20% in the mean and 13% in the median)
- With wind speeds lower 1 m/s emission rates are underestimated by 70%
- Mean estimated emission rate differs in average $34 \pm 47\%$ and median $38 \pm 42\%$ to the actual release rate (without Zero- and 0.01-Release)
- Some measurements are slightly influenced by other emitters nearby
- Enhancements up to 950 ppb were measured at zero release, appearing to originate from the released area